

CASA0023 Remotely Sensing Cities and Environments Learning Diary

Christy Choi

2026-02-01

Table of contents

Preface	3
About me,	3
In this Learning Diary,	3
1 Introduction	4
1.1 Summary	4
1.1.1 Types of Sensor	4
1.1.2 Types of Resolution	4
1.1.3 Study Area: Lisboa, Portugal	5
1.2 Application	8
1.3 Reflection	8
1.4 References	8
2 Summary	9
References	10

Preface

Hello and welcome to Christy's page on Remote Sensing. This page is an online learning diary for CASA Remotely Sensing Cities and Environments 25/26. This website contains a record of her week-by-week learnings, including summary of key concepts, practical applications, reviews on academic literature and relevant web sources, as well as reflections and discussions on learnt concepts and techniques.

About me,

She is a current master's student at the Bartlett Centre for Advanced Spatial Analysis (CASA), University College London (UCL) reading MSc, Urban Spatial Science. She holds a bachelor's degree in BSc, Planning and Real Estate from the University of Manchester (2022-2025).

Email: chi.choi.25@ucl.ac.uk

Github: [christychoicc](https://github.com/christychoicc)

In this Learning Diary,

Each chapter will be covering contents taught in the corresponding week:

1. Introduction
 - Getting started with remote sensing, introductions, sensors, spatial resolutions and practical pre-processing
2. Sensor

1 Introduction

Remote sensing is not only satellite imagery but a system that allows us to obtain EO data through acquiring information from a distance using sensors on platforms. This section introduces types of sensors and resolutions of remote sensing data. The following study area of Lisboa, Portugal will demonstrate a workflow of analysing Earth Observations (EO) data processed through QGIS and SNAP.

1.1 Summary

1.1.1 Types of Sensor

Passive Sensors detect energy emitted or reflected from the object by the sun. They mainly operate in the visible, infrared, thermal infrared, and microwave bands of the electromagnetic spectrum, mostly used to measure surface temperature, vegetation properties and other physical attributes. However, passive sensors cannot penetrate dense cloud cover, limiting areas with higher cloud rates.

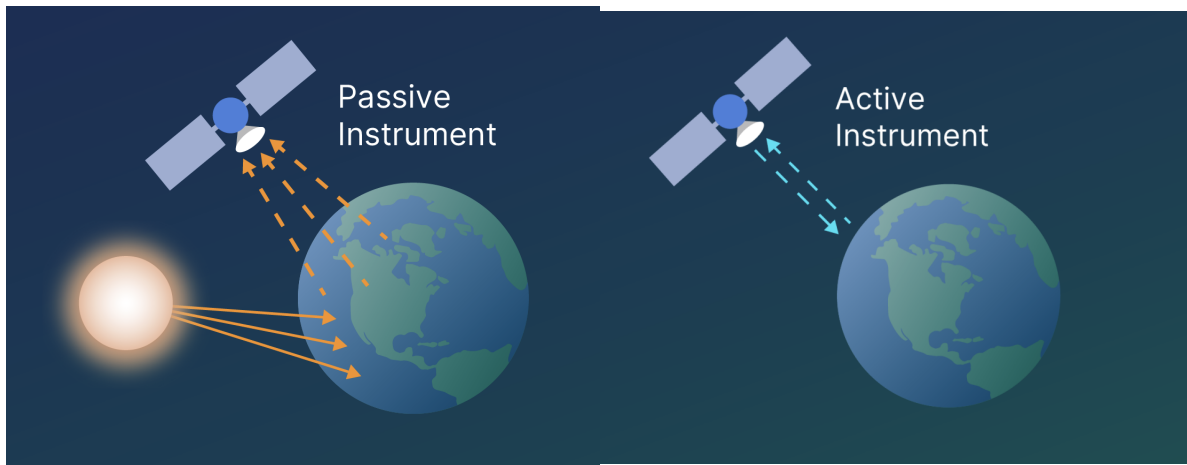
Active Sensors emits energy and collect data based on changes in the return signal. Most active sensors have higher capabilities to penetrate atmospheric conditions. For example, the Synthetic Aperture Radar (SAR), has the ability to “see through cloud” as it operates with comparably longer wavelengths, allowing data collections during night hours. This allows them to be less dependent on sunlight which better handles atmospheric constraints.

Overall, observations are shaped by the electromagnetic spectrum and its interactions between radiation, atmosphere, and Earth’s surface, where raditaion can be absorbed, transmitted, or scattered (Tempfli *et al.*, 2009).

1.1.2 Types of Resolution

1.1.2.1 Spatial Resolution

The size of the raster grid cell per pixel, where it can range from centimetres to kilometres (e.g. 20cm, 30m or 1km)



(a) Passive Sensor (Source: NASA EARTHDATA) (a) Active Sensor (Source: NASA EARTHDATA)

1.1.2.2 Spectral Resolution

Visible part of the electromagnetic spectrum is composed of red, green and blue wavelengths. Multispectral sensors measure reflection at multiple wavelength bands to create spectral signatures, which such patterns of reflectance across the spectrum helps distinguish different surface features. However, observations are often constrained to atmospheric windows (e.g. cloud cover) which absorbs radiation.

1.1.2.3 Temporal Resolution

The time it revisits

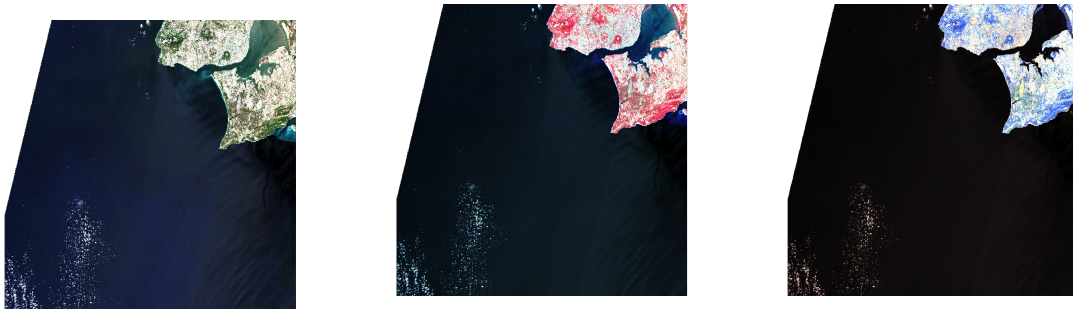
1.1.2.4 Radiometric Resolution

1.1.3 Study Area: Lisboa, Portugal

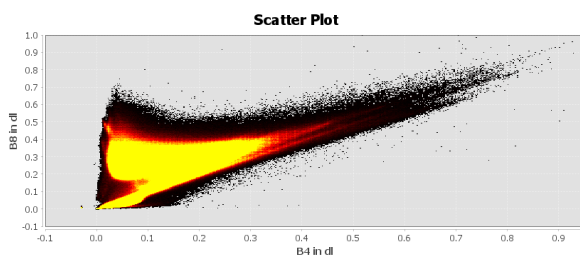
Pre-processing in SNAP, used both Sentinel and Landsat.

1.1.3.1 Tasseled Caps

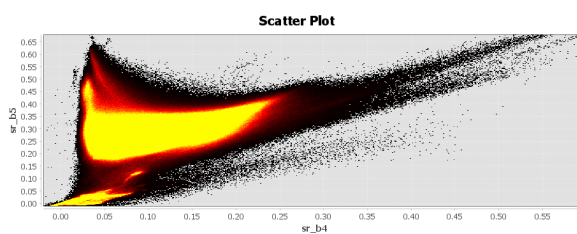
Maths Band Equations:



(a) True Natural Colour, Sentinel (a) False Colour Composite, Sentinel (a) Atmospheric Penetration, Sentinel



(a) Sentinel Band 4 and Band 8 Scatterplot

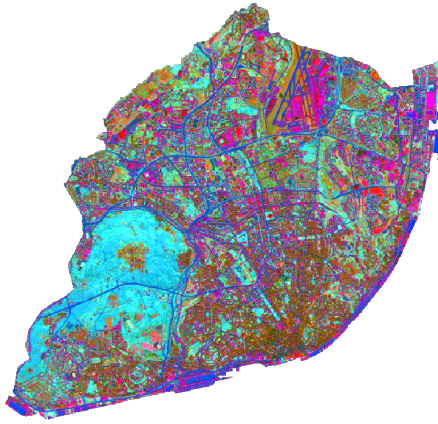


(a) Landsat Band 4 and Band 5 Scatterplot

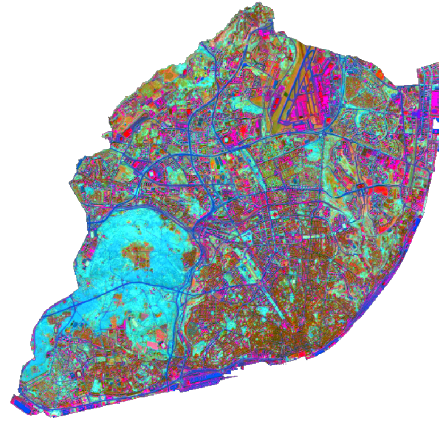
$$\begin{aligned} \text{Brightness} = & 0.3037(B2) + 0.2793(B3) + \\ & 0.4743(B4) + 0.5585(B8) + \\ & 0.5082(B11) + 0.1863(B12) \end{aligned}$$

$$\begin{aligned} \text{Greenness} = & -0.2848(B2) - 0.2435(B3) \\ & -0.5436(B4) + 0.7243(B8) + \\ & 0.0840(B11) - 0.1800(B12) \end{aligned}$$

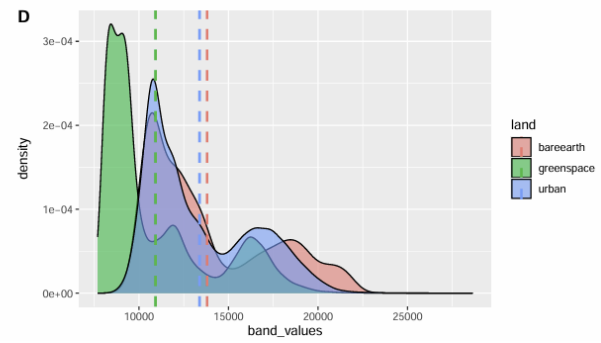
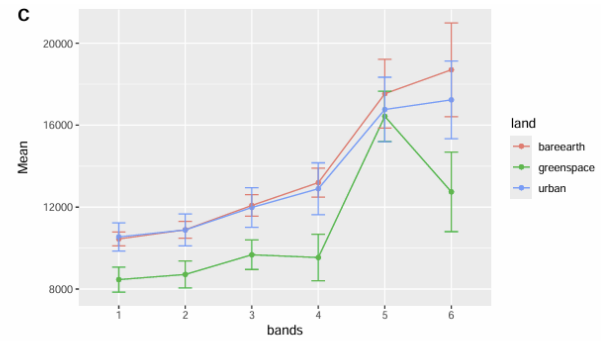
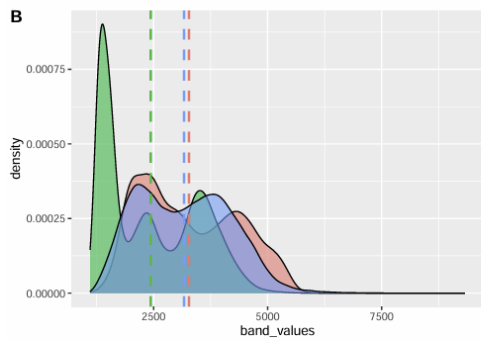
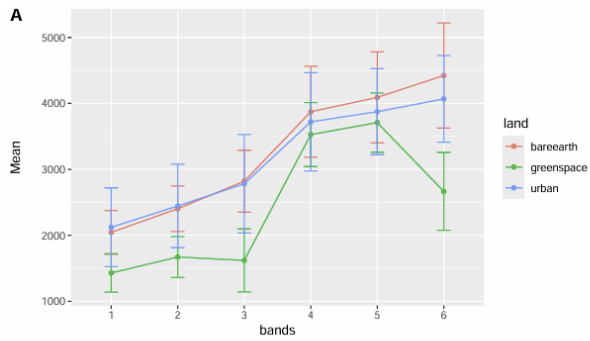
$$\begin{aligned} \text{Wetness} = & 0.1509(B2) + 0.1973(B3) \\ & +0.3279(B4) + 0.3406(B8) \\ & -0.7112(B11) - 0.4572(B12) \end{aligned}$$



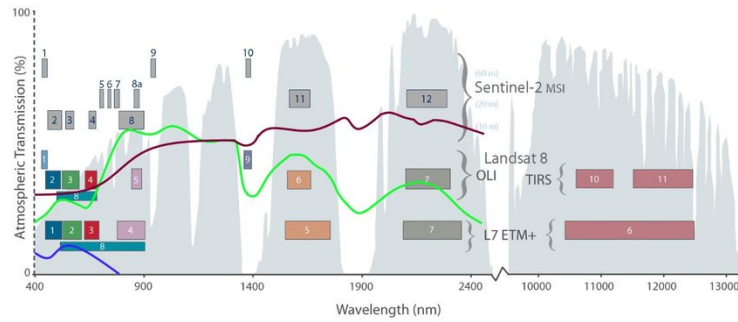
(a) Lisboa in August 2025 (Sentinel, Tasseled Cap Transformed)



(a) Lisboa in December 2025 (Sentinel, Tasseled Cap Transformed)



(a) Spectral Reflectance of Sentinel (A, B) and Landsat (C, D)



(a) Spectral Bands of Sentinel 2 and Landsat 7 8

1.2 Application

1.3 Reflection

1.4 References

Sentinel Data Source:

https://identity.dataspace.copernicus.eu/auth/realms/CDSE/protocol/openid-connect/auth?client_id=sh-5f8b630b-b083-49ed-b340-b8f01ecb81c4&redirect_uri=https%3A%2F%2Fbrowser.dataspace.copernicus.eu%2F%3Fzoom%3D5%26lat%3D50.16282%26lng%3D20.78613%26demSource%3D%3D%2522MAPZEN%2522%26cloudCoverage%3D30%26dateMode%3DSINGLE&state=26c991f3-8c5e-487e-b23d-d5fe36fbabb7&response_mode=fragment&response_type=code&scope=openid&nonce=c67a4727-3008-4473-8b84-0a9304c87ce3&code_challenge=ZGmOypxnXCivN18EjxS722lF4SDVSsskdcUb_zktYp4&code_challenge_method=S256

2 Summary

In summary, this book has no content whatsoever.

1 + 1

[1] 2

References